Automatic Question Answer Generation Based on Gujarati Paragraph

PhD Weekly Progress Presentation — 2

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Work Completed

- Completed Gujarati Question Answer dataset creation
- Structured dataset in JSON format
- Performed preprocessing and validation
- Conducted initial dataset analysis
- Verified linguistic correctness of questions and answers

Gujarati QA Dataset Overview

- Dataset is created from Gujarati paragraph-based educational content
- Covers multiple subject domains:
  - History
  - Literature
  - Wildlife
  - Religion
- Dataset designed for extractive Question Answering tasks
- Questions aligned with educational learning objectives
- Suitable for training Small Language Models (SLMs)

Dataset Statistics

Metric	Value
Total Context Paragraphs	7
Total Question–Answer Pairs	190
Average QA per Context	27
Language	Gujarati
Format	JSON
Answer Type	Extractive Span

Dataset Characteristics

- Total dataset size: **190 questions**
- 100% Gujarati linguistic content with natural interrogative sentence structures
- Multiple question categories based on Bloom's taxonomy (**10 types**)
- Balanced difficulty distribution:
  - Easy: **70 questions (36.8%)**
  - Medium: **61 questions (32.1%)**
  - Hard: **59 questions (31.1%)**
- Short answer spans suitable for extractive QA models
- Educationally relevant literary content for learners

Question Type Coverage

- Factual (Basic Recall): **31 questions (16.3%)**
- List (Enumeration): **27 questions (14.2%)**
- Inferential (Reasoning): **24 questions (12.6%)**
- Numerical/Date: **17 questions (8.9%)**
- Comparative: **14 questions (7.4%)**
- Definition: **13 questions (6.8%)**
- Evaluative: **13 questions (6.8%)**
- Thematic: **13 questions (6.8%)**
- Predictive: **12 questions (6.3%)**
- Complex/Mixed: **26 questions (13.7%)**

Dataset Example

Context

અમદાવાદ ગુજરાત રાજ્યનું સૌથી મોટું શહેર છે. તેની સ્થાપના સુલતાન અહમદ શાહે કરી હતી.

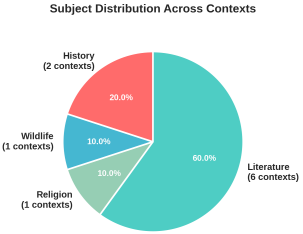
Question

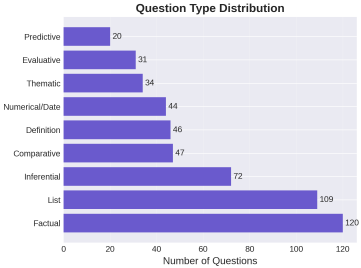
અમદાવાદની સ્થાપના કોણે કરી હતી?

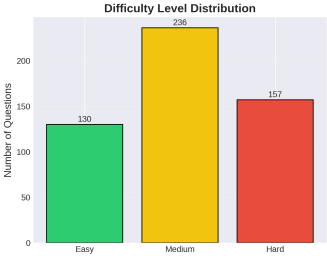
Answer

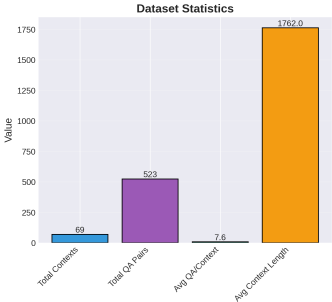
સુલતાન અહમદ શાહે

Dataset Analysis Visualization









Work Plan for Next Week

- Prepare dataset for model training
  - Tokenization and preprocessing
  - Train–validation split
  - Input formatting for QA models
- Train baseline Extractive Question Answering model
  - Fine-tuning transformer-based architecture
  - Initial hyperparameter selection

Thank You

Questions?

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